Received: 2 June 2016 | Revised: 1 February 2018 | Accepted: 1 February 2018
DOI: 10.1002/rob.21782

REGULAR ARTICLE

WILEY

The AutoSOAR autonomous soaring aircraft, part 1: Autonomy algorithms

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Funding information
Office of Naval Research, Grant/Award Number: N000141110656

Abstract

Autonomous soaring has the potential to greatly improve both the range and endurance of small robotic aircraft. This paper describes an autonomous soaring system that generates a dynamic map of lift sources (thermals) in the environment and uses this map for online flight planning and decision making. Components of the autonomy algorithm include thermal mapping, explore/exploit decision making, navigation, optimal airspeed computation, thermal centering control, and energy state estimation. A finite state machine manages the aircraft behavior during flight and determines when changing behavior is appropriate. A complete system to enable autonomous soaring is described with special attention paid to practical considerations encountered during flight testing. A companion paper describes the hardware implementation of this system and the results of a flight test campaign conducted at Aberdeen Proving Ground in September 2015.

KEYWORDS
aerial robotics, bioinspired methods, exploration, mapping, planning

1 | INTRODUCTION

Autonomous soaring has now been the subject of research for nearly two decades. Since the problem was first addressed in detailed simulations (Wharington, 1998), several autonomous soaring UAVs have been developed and subsequently demonstrated. The first such successful demonstration occurred in the summer of 2005 following numerous feasibility surveys in a simulated environment (Allen & Lin, 2007). In 2008, a fully autonomous aircraft showed impressive endurance and range while competing successfully against human piloted competitors in the Montague Cross Country Challenge for remote-controlled sailplanes (Edwards & Silverberg, 2010). Recently, researchers have demonstrated cooperative operation of soaring aircraft with promising results (Andersson, Jones, Dobrokhodov, & Kaminer, 2012a; Hazen, 2012).

The reports of such successful flights are notably fewer in number than those regarding simulated investigations. The typical approach to simulation employs a regularly shaped thermal model with a circular cross section and a strength that is greatest in the center and decreases radially (Allen, 2006). Commercial flight simulation packages with advanced (but not openly shared) atmospheric models have also been used to demonstrate developments (Camacho, Dobrokhodov, & Jones, 2014; Cowling, Willcox, Patel, Smith, & Roberts, 2009; Daugherty & Langelaan, 2014).

Because these simulations have relatively simple environmental models, many developments which show promise in simulation are not robust enough for flight in a realistic environment. For example, thermals often have inconsistent and irregular shapes (Andersson, Kaminer, Dobrokhodov, & Cichella, 2012b), a fact that was observed during our flight tests and that makes energy extraction more difficult. Methods have been developed to account for this by fitting models to observed thermals in flight (Edwards, 2008a), using UKFs (Hazard, 2010), particle filters (Bencatel, 2010), and two-dimensional splines (Bird & Langelaan, 2012), but not all of these approaches have been flight-tested.

Even with small autopilots and computers available, realizing an autonomous soaring system is challenging as several problems must be solved before an aircraft can be fielded. First, an effective soaring system must be able to sense the surrounding environment and the motion of atmospheric currents. Second, the aircraft must be equipped to make decisions to exploit energy and avoid sinking air. Third, the aircraft must consider its energy state and that of the environment as it navigates.

This paper describes the design of the AutoSOAR (Autonomous Sailplane for Atmospheric Research) system, a nearly complete set of autonomous soaring algorithms intended to be flown on an appropriate aircraft. "Nearly complete" means that autonomous take-off and landing are not included, but all phases of *soaring* flight (thermal detection, thermal latching and delatching, thermal centering control, mapping, exploration, and flight management) are addressed. The novel contributions of the work presented here are

- energy state estimation and mapping of energy sources in the environment (denoted "atmospheric map"—this atmospheric map comprises a wind map, a lift feature map and a lift occupancy map);
- control and guidance of flight based on the wind map (calculation of the reachable space based on the thermal map, terrain, and winds) and on the lift feature map (evaluation of when mapped lift sources should be exploited and which mapped lift sources should be exploited);
- algorithms for exploration based on the atmospheric map and on heuristic predictions of where thermals are likely to occur based on terrain, solar incidence, ground cover, and where thermals have been previously found.

Other components (e.g., thermal centering control, thermal latching, and unlatching) are adapted from previous literature and are covered here for completeness. In all cases, a strong focus on implementing algorithms on a small single-board computer (here, an ODROID-XU4) drove algorithm design to easily implementable, low computational overhead methods. Where appropriate, flight data are used to show the utility of components of the soaring system. Note that all algorithms were supported by extensive simulation (both in MatLab and hardware in the loop using the simulation environment supplied with the Piccolo autopilot). In the interest of space, these simulations are not discussed in detail here, but references to papers describing earlier simulation results are given in the appropriate parts of this paper. More detailed flight data (along with a description of hardware implementation) are given in the companion paper (Depenbusch, Bird, & Langelaan, 2017).

Autonomous soaring systems often feature aircraft following heuristic rules, which are typically based on the tactics employed by human pilots. Many such rules are utilized in our work including optimal flight speed selection (Cochrane, 1999b; MacCready, 1958), preferential exploration of regions with promising features (Pagen, 1992; Reichmann, 1978), and thermal climbing strategies. Of particular interest to researchers have been the strategies by which the climb rate of an aircraft is maximized when in thermal lift. These include opportunistically changing turn direction during a climb (Akos, Nagy, Leven, & Vicsek, 2010), feedback control on measured changes in lift strength (Allen & Lin, 2007; Andersson & Kaminer, 2009), and extremum seeking methods to identify the optimal orbit radius around an estimated thermal center (Daugherty & Langelaan, 2014).

Techniques developed for mapping the convective boundary layer range from grid-based methods (Bower, Flanzer, Naiman, & Saripalli, 2010; Depenbusch & Langelaan, 2011) to Gaussian process regression (Lawrance & Sukkarieh, 2011). Information is introduced into these maps from measurements taken by soaring aircraft during flight. These approaches have relied primarily on the uncertainty in a map of the wind to guide exploration behavior. More complex exploration strategies, such as seeking regions of high solar incidence, have been examined in simulations (Andersson, Kaminer, Jones, Dobrokhodov, & Lee, 2009; Cheng & Langelaan, 2014). Earlier work by the authors has demonstrated via simulation that even rudimentary maps improve endurance of a soaring aircraft by a factor of two (and cooperative mapping by a team of soaring aircraft improves endurance significantly more) (Cheng & Langelaan, 2014; Depenbusch & Langelaan, 2011) Recently, exploration behaviors that minimize uncertainty in a map of the environment have been further refined (Chung, Lawrance, & Sukkarieh, 2014).

With some knowledge of the environment and uncertainty, strategies for planning paths have been developed (Chakrabarty & Langelaan, 2011). Applications have been developed where thermal soaring and energy exploitation have been integral to the mission (Acevedo, Lawrance, Arrue, Sukkarieh, & Ollero, 2014; Cobano, Alejo, Vera, Heredia, & Ollero, 2013; Cutler, McLain, Beard, & Capozzi, 2010; Makovkin & Langelaan, 2014). Missions that an autonomous soaring aircraft would be well suited for include persistent surveillance, observation of atmospheric phenomena (Zhang, 2012), and to bolster communication networks (Andersson et al., 2012a).

2 | SYSTEM OVERVIEW

The developments that follow cover the high-level components of an autonomous soaring system. In this work, the aircraft is assumed to be equipped with an autopilot acting as a low-level vehicle controller whereas the AutoSOAR system runs on a separate onboard computer (the "payload" computer).

Figure 1 presents a diagram of the AutoSOAR system. Within this architecture, the autopilot (blue boxes) provides a stream of sensor information to the payload computer, which runs the soaring system. Additionally, the autopilot executes low-level control of the vehicle by actuating control surfaces to track outer-loop commands (airspeed, bank angle, destination waypoint, etc.). The outer-loop commands originate exclusively from the payload computer when the autonomous soaring system is active.

The challenges faced by an autonomous soaring aircraft require numerous systems to simultaneously analyze and interpret data. Because of this, the vehicle software uses the robot operating system (ROS) (Quigley et al., 2009), which facilitates exchange of information between distributed processes. Each of the functional components is implemented on the flight computer as a ROS node. As an example, separate nodes communicate with the autopilot, estimate the aircraft energy state, and construct maps of the environment. A simplified diagram of all system processes is depicted by the green boxes in Figure 1.

The energy state computation block computes aircraft total energy and its first two derivatives as well as estimates of vertical and horizontal wind speed (Section 3.2). All components subscribe to the published energy and energy rate data. The atmospheric mapping block uses estimates of vertical wind and energy rates to generate a map of vertical wind speed (Section 3.3). This wind map is processed to calculate the glide footprint (Section 4.1) and to identify useful lift (Section 3.4). The lift feature map is used to compute optimal speed to fly using a variant of MacCready speed to fly (Section 4.2); it is used in

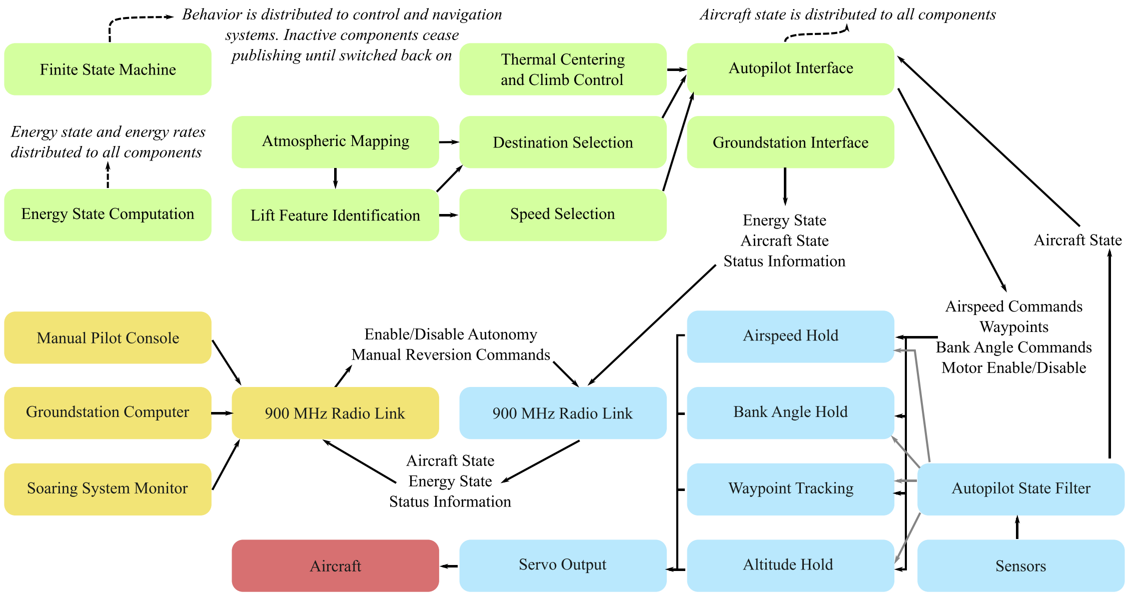


FIGURE 1 System diagram for the AutoSOAR autonomous soaring system. Boxes in blue show autopilot functions. Green boxes show the on-board autonomy system. Yellow boxes indicate the ground system used for monitoring only. Arrows indicate the flow of information through the system. Dashed lines and italicized fonts indicate conceptual, rather than direct messages, simplified for clarity

combination with the atmospheric map to select destinations for exploration (Section 5.1) or exploitation of vertical air motion (Section 5.2). When a thermal is encountered (either because it was deliberately sought or because it was randomly encountered), the thermal centering control seeks to maximize climb rate by estimating the center of the thermal, determining the turn radius that maximizes climb rate, and following the resulting trajectory (Section 4.3).

In all, the AutoSOAR system involves the interaction of 18 separate nodes. While there is overhead associated with running 18 ROS nodes, the resulting modularity means that various subcomponents are easily switched and modified. Both hardware in the loop tests and flight tests showed that the overhead was manageable on the payload computer (an ODROID-XU4).

The hardware separation of the flight controller and the computer engaged in high-level autonomous soaring calculations allows control of the aircraft to be maintained regardless of the system load on the payload computer. This frees the soaring computations from hard real-time requirements. Additionally, the autonomous soaring system is made agnostic to the low-level controller used (provided it can follow a basic set of commands), further increasing its applicability to other soaring aircraft.

3 | PERCEPTION AND MAPPING

3.1 | Aircraft performance model

The speed polar for the aircraft is a simple yet useful description of the aircraft's aerodynamic performance over a range of flight speeds. As shown in Figure 2, the speed polar relates airspeed to expected sink rate in still air and in steady flight. A speed polar is typically generated by taking measurements of the aircraft sink rate across a range of

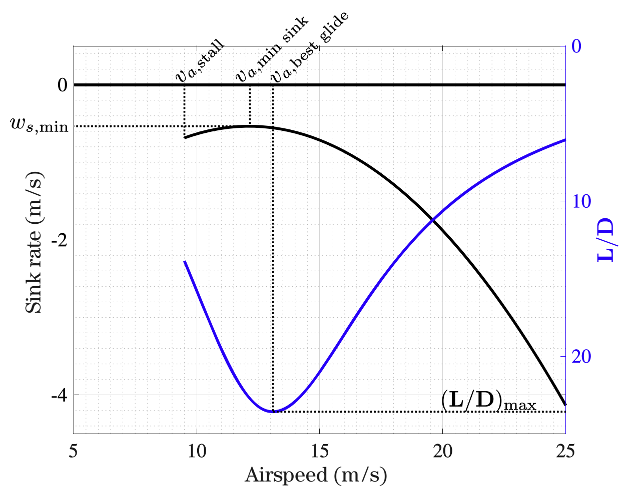


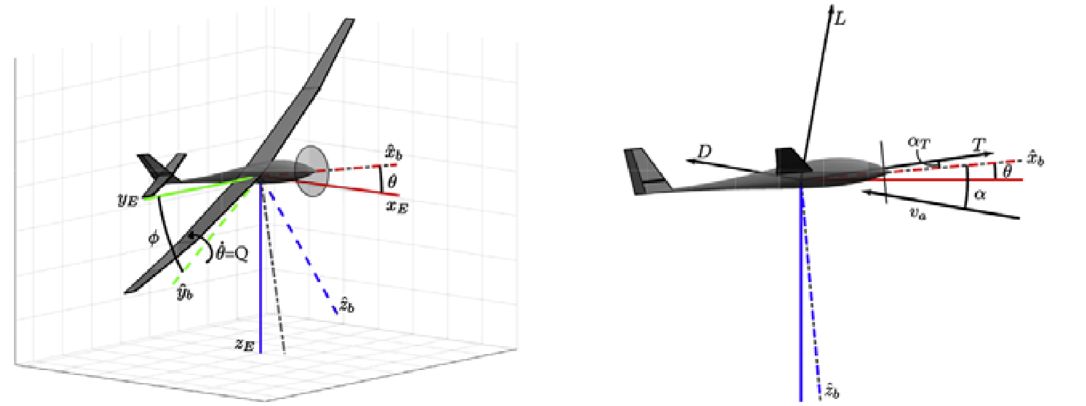
FIGURE 2 Performance diagram for RnR Products SB-XC. This speed polar (black curve) approximates the relationship between airspeed and the sink rate of the aircraft. The lift to drag ratio is a measure of the glide efficiency of the aircraft and is shown in blue

airspeeds (Edwards, 2008b) (the speed polar and *L*/*D* curves for the SB-XC aircraft shown in Figure 2 were generated from a combination of data in Edwards (2008b) and from AutoSOAR flight data).

A quadratic function of airspeed is often used to approximate speed polar over a nominal speed range (Reichmann, 1978).

$$w_s\left(v_a\right) = a\,v_a^2 + b\,v_a + c \tag{1}$$

This low-order model will be used repeatedly in the following development to capture the expected performance of the aircraft. Note that in steady-state (constant airspeed) gliding flight, the sink rate is exactly equal to the rate of change of specific total energy (i.e., specific power).



(a) The aircraft is shown with a positive pitch θ and roll ϕ orientation.

(b) The relevant longitudinal states are shown. Lift and drag are defined perpendicular and parallel to the airspeed vector. Thrust offset from the body-axis is defined by α_T.

FIGURE 3 Definition of aircraft states and conventions relevant to the energy-state estimation problem

The wings-level speed polar presented in Equation 1 can be modified to account for the additional power lost during maneuvering flight leading to a model for drag in terms of specific energy rate. When an aircraft is in a steady banked turn ($|\phi| > 0$), the wing must produce lift as though it were flying at $v_a\sqrt{\cos\phi}$ (Edwards et al., 2016; Zhang, 2012). This, in addition to the increase in power lost to drag during a banked turn, yields the following energy-rate expression:

$$\dot{e}_{\text{drag}}(v_a,\phi) = \cos^{-3/2}\phi\left[a\left(v_a\sqrt{\cos\phi}\right)^2 + b\left(v_a\sqrt{\cos\phi}\right) + c\right] \tag{2}$$

3.2 | Energy state and wind estimation

Total energy and the first two time derivatives of total energy are useful in sensing atmospheric conditions, and an accurate and low-latency estimate of total energy is critical to successful implementation of a soaring system (Andersson et al., 2009; Depenbusch & Langelaan, 2011; Daugherty & Langelaan, 2014).

Because forces acting on the aircraft arise from a dynamic reaction with the air, the airspeed determines the useful energy of the aircraft. The aircraft's total air-relative specific energy (i.e., total energy normalized by aircraft weight) and its time derivatives are

$$e = \frac{v_a^2}{2g} + h \tag{3}$$

$$\dot{e} = \frac{v_a\dot{v}_a}{g} + \dot{h} \tag{4}$$

$$\ddot{e} = \frac{\dot{v}_a^2 + v_a\ddot{v}_a}{g} + \ddot{h} \tag{5}$$

where v_a is airspeed as defined in Figure 3, and h is the height above a datum. Specific energy conveniently has units of altitude. Because the total energy and its derivatives cannot be directly measured, an estimator is used to fuse sensor measurements. An unscented Kalman filter (UKF) is used to estimate the energy state:

$$\mathbf{X} = \left[e, \dot{e}, \ddot{e}, v_a, \dot{v}_a, \alpha\right]^T \tag{6}$$

A UKF is accurate to roughly third order, whereas the extended Kalman filter is accurate to only first order and has similar computational complexity (Wan & van der Merwe, 2000).

Assuming that the sideslip angle is small, which is reasonable for an aircraft with closed-loop rudder control, and that the aircraft is flying in constant wind, the state dynamic equations are

$$\frac{\partial}{\partial t}\begin{bmatrix} e \\ \dot{e} \\ \ddot{e} \\ v_a \\ \dot{v}_a \\ \alpha \end{bmatrix} = \begin{bmatrix} \dot{e} \\ \ddot{e} \\ \ddot{e} \\ \frac{T\cos(\alpha+\alpha_T) - D}{m} + g(-\cos\alpha\sin\theta + \sin\alpha\cos\phi\cos\theta) \\ \ddot{v}_a \\ Q - \frac{T\sin(\alpha+\alpha_T)+L}{v_a m} + \frac{g}{v_a}(\sin\alpha\sin\theta + \cos\alpha\cos\phi\cos\theta) \end{bmatrix} \tag{7}$$

Measurements of aircraft states are related to this system by the equations:

$$\begin{bmatrix} h \\ \dot{h} \\ \ddot{h} \\ v_a \\ a_z \end{bmatrix} = \begin{bmatrix} e - \frac{v_a^2}{2g} \\ \dot{e} - \frac{v_a\dot{v}_a}{g} \\ \ddot{e} - \frac{\dot{v}_a^2 + v_a\ddot{v}_a}{g} \\ v_a \\ \frac{-L}{mg} \end{bmatrix} \tag{8}$$

It is assumed in this estimator that $\ddot{v}_a$ and $\ddot{e}$ are small during the typical maneuvers performed by the vehicle in flight. The autopilot accelerometer measures a_z, acting along the body z-axis $\hat{z}_b$ directly, while $\ddot{h}$ is the projection of the accelerometer measurements onto the inertial z-axis z_E. Altitude and vertical rate (h and $\dot{h}$) as well as the input vector $[\phi, \theta, Q, \delta t, \delta f]$ are all readily available from the autopilot. Lift, drag, and thrust forces are computed by a low-order aircraft model using stability derivatives (Stevens & Lewis, 2003).

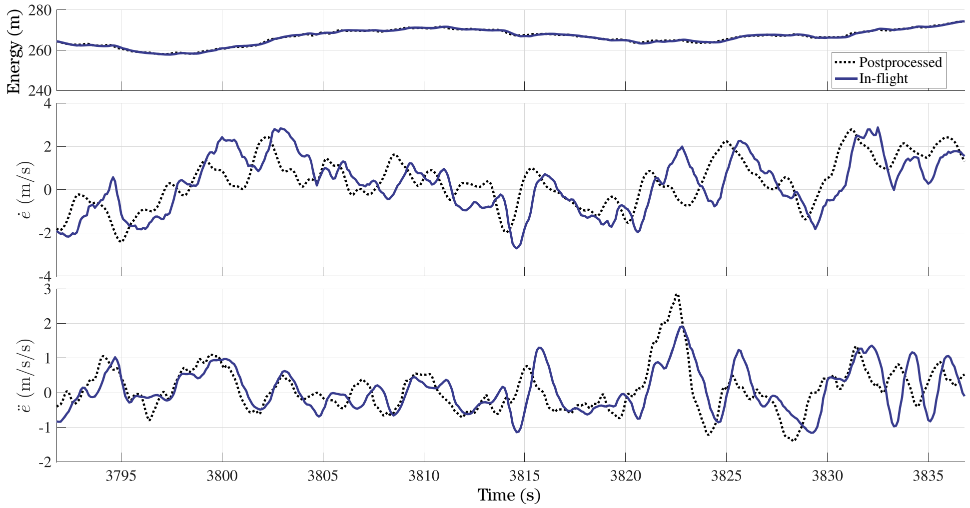


FIGURE 4 The output of the energy-state estimator for a typical flight segment, comparing in-flight estimated and postprocessed energy state. Delay in the estimated energy rate and energy acceleration are evident when comparing the postprocessed energy states with those estimated in flight

The UKF is implemented following the algorithm described in Wan and van der Merwe (2000), with $\alpha = 1$, $\beta = 2$, and $\kappa = 0$. Figure 4 compares the output of the online energy estimator to a postprocessed estimate (which uses future aircraft states for lag-free smoothing). This postprocessed energy state was developed by estimating the values of $h, \dot{h}, \ddot{h}$ and sensor biases with a Rauch–Tung–Striebel Kalman smoothing filter from measured GPS altitude, GPS velocity, and accelerometer data. Airspeed was smoothed and its derivative obtained by a zero-phase Savitsky–Golay filter second-order polynomial fit to a 1.5-s long window). The energy state was then determined directly using Equations 3–5.

Delay between aircraft energy changes and recognition of those changes by the soaring system has been reported as a problem by other autonomous soaring researchers (Allen & Lin, 2007; Andersson et al., 2012b). Reported delays of up to 3 s in energy-state measurement reduced the performance of feedback control while thermalling and made precisely locating lift features difficult (Andersson et al., 2009).

To approximate the time delay in the filter, the peak of the cross-correlation between the output of the UKF recorded in flight and the postprocessed energy state was identified (Hero et al., 1998). No measurable delay is evident in the energy estimate, 0.57 s of delay is present in the energy-rate estimate, and the energy-acceleration estimate is delayed by 1.13 s. At flight speeds typically used during soaring by the AutoSOAR aircraft (12.5–15 m/s), the time delay in energy rate estimate corresponds to 7–8.5 m flight distance. At a typical thermalling radius of 40 m, this corresponds to a phase delay of approximately 12°. Flight testing as well as earlier simulation work (Andersson et al., 2009; Daugherty & Langelaan, 2014) suggests that this is adequate for good thermalling performance.

If appropriate sensors are available, wind speed can be computed as the difference between inertial and air-relative velocity of the aircraft (Langelaan, Alley, & Neidhoefer, 2011),

$$\mathbf{v}_g = \mathbf{v}_a + \mathbf{v}_w \tag{9}$$

where $\mathbf{v}_g$ is the ground-relative (inertial) velocity of the aircraft and the subscripts a and w indicate air-relative and wind velocities, respectively. The inertial velocity of the aircraft can be obtained from GPS with high precision. Wind-relative velocity is more difficult to obtain using the sensors available on a standard small aircraft autopilot: airspeed is measured using the pitot-static system, but angle of attack and angle of sideslip sensors (vanes or specialized pressure taps) are generally not available on a small aircraft.

Here it is assumed that both angle of sideslip and angle of attack are small (closed-loop rudder control results in small sideslip angle, and angle of attack cannot exceed 10–15 deg before the onset of stall, hence $\cos\alpha \approx 1$). Under these conditions, the horizontal components of wind speed can be computed from Equation 9.

However, even small errors in assumed α lead to large errors in computed vertical wind speed. The vertical wind is instead determined by subtracting the anticipated sink rate from the estimated aircraft energy-rate $\hat{\dot{e}}$ presented in Section 3.2. If the aircraft sinks more quickly than expected, it must be flying through sink and vice versa in lift. Assuming a shallow glide angle (large L/D) and steady-state flight, the vertical wind speed (positive downwards) may be obtained by

$$w_z = -\hat{\dot{e}} + \dot{e}_{\text{drag}} - \dot{e}_{\text{motor}} \tag{10}$$

where $\dot{e}_{\text{drag}}$ is the expected sink rate from Equation 2 (Zhang, 2012), corrected for turning flight. Assuming that the errors in the estimated quantities of Equation 10 have Gaussian distribution, the variance in vertical wind speed is

$$\sigma^2_{w_z} = \sigma^2_{\hat{\dot{e}}} + \sigma^2_{\dot{e}_{\text{drag}}} + \sigma^2_{\dot{e}_{\text{motor}}} \tag{11}$$

Here

$$\sigma^2_{\dot{e}_{\text{drag}}} \approx n^3 \operatorname{var}(w_s) \tag{12}$$

where $n = \frac{1}{\cos\phi}$ is load factor. Hence accurate measurements of vertical wind are difficult to obtain during maneuvering flight (when load factor is high). It should be noted that errors in the speed polar will directly correspond to errors in the vertical wind speed measurement as shown in Equation 10. Practically, errors in the polar are accounted for by inflating the variance of that component, var(w_s). If an adequate model for motor power as a function of throttle setting and propeller efficiency (itself a function of airspeed and throttle setting) is available, vertical wind speed can be estimated during powered flight as well as during gliding flight. In our implementation, flight operation consisted of a short duration climb at maximum throttle followed by gliding while mapping and searching for thermals. During the maximum throttle climb, the vertical wind estimator is disabled.

3.3 | Atmospheric mapping

Many planning algorithms assume that the environment is known (Chakrabarty & Langelaan, 2009). For soaring flight, knowledge is both incomplete and uncertain. Meteorological models can provide information about expected mean wind speeds and the likelihood of thermal activity over a large region (Glendening, 2002; Hindman, Saleeby, Liechti, & Cotton, 2007), but cannot give real-time winds or the precise locations of thermals. Knowledge of updraft locations could enable traditional path planning algorithms to approach the problem of optimal interthermal flight (Chakrabarty & Langelaan, 2013).

To address this, an atmospheric energy map is developed to represent the aircraft's belief in the state of the wind. The map combines wind measurements taken by the aircraft with a model of how wind features evolve with time so that the precise locations of phenomena such as thermals can be identified. This map is built with the assumption that the aircraft operates within a confined region so that it can reencounter updraft features. In prior work (Cheng & Langelaan, 2014; Depenbusch & Langelaan, 2011), a map of updraft locations enhanced the capability of aircraft to exploit environmental energy during simulated missions.

A useful wind map should include both an estimate of the magnitude of the wind and the uncertainty in the estimate. Bower et al. (2010) present an occupancy grid inspired approach to thermal mapping but do not address uncertainty in either measurements or the map. The map presented in our work (1) tracks uncertainty in the expected vertical component of wind in addition to the magnitude, (2) is scalable to incorporating the measurements from multiple aircraft, and (3) can readily be modified to incorporate more complex models of the environment.

To construct the map, the environment is divided into $M \times N$ uniform two-dimensional cells that cover the mission space. The size of a cell is small enough that the spatial variation of wind over the cell is small, but large enough that small-scale variations (e.g., gusts, turbulence) average to zero over one cell.

3.3.1 | Dynamic model of thermals

Thermals decay in roughly one eddy turnover time—the boundary layer depth divided by the convective velocity scale (Stull, 1988). Conservation of mass requires that mean vertical wind over large areas and longer periods of time (i.e., large enough and long enough to encompass several thermals) is near zero. The vertical wind speed in a cell can thus be modeled as an exponential decay:

$$\hat{w}_{z,ij,k+1} = A\hat{w}_{z,ij,k} + \mathcal{N}(0, Q) \tag{13}$$

where the scalar A is chosen in [0, 1] such that the wind speed decays to a small fraction of its original value within one thermal lifespan. $\mathcal{N}(0, Q)$ represents zero-mean Gaussian process noise, with Q scaled such that the confidence in an estimate erodes over a similar timespan.

For a given cell, this relaxation model can be computed independently of other cells. By assuming that process noise is spatially uncorrelated, the mapping problem can be cast as a set of $M \times N$ 1D Kalman filters.

3.3.2 | Measurement model

Measurements of vertical wind speed are used to drive the measurement update step of each cell's vertical wind speed estimate. In addition to on-board measurements, vertical wind speed data could come from sources such as other aircraft, balloons, or LIDAR. If several aircraft are cooperating, maps can be generated independently by each aircraft, reducing communication overhead by sharing only measurements rather than the entire map.

It is assumed that each wind measurement is independent and corrupted by Gaussian noise with variance $R_{ij,k}$.

$$R_{ij,k} = \sigma^2_{w_{z,k}} + \sigma^2_{\text{gust}} + \sigma^2_{d(k,ij)} \tag{14}$$

The quality of the wind estimate (discussed in Section 3.2) is included in the term $\sigma^2_{w_{z,k}}$. Wind measurements also include gust features too small to be utilized by the aircraft so the variance is further increased by σ_{gust}, defined by the Dryden turbulence model (Hoblit, 1988). The term $\sigma^2_{d(k,ij)}$ is a localization term used to spread a measurement into surrounding cells. The localization is defined:

$$\sigma^2_{d(k,ij)} = \max\left(\left(\frac{d(k,ij)}{r_{max}}\right)^2, 1.0\right) \tag{15}$$

where d represents the distance between the measurement location and the center of cell ij, and r_{max} is the nominal thermal radius, 50 m is used in this work (Cheng & Langelaan, 2014; Depenbusch & Langelaan, 2011).

Clearly, spatial correlation in the wind map would arise because of this measurement model. To improve computational tractability, this correlation is ignored so that both measurement and time updates for the wind estimate involve only scalar computations. Measurements are applied to the map as they are generated by the estimator (10 Hz) while time updates are computed at 1 Hz.

An example of a section of the vertical wind map generated during a flight is shown in Figure 5. Measurements of vertical wind speed are illustrated by the flight path coloring and registered with the cells immediately surrounding point of measurement. When the aircraft circles in an updraft, many measurements are applied to a small region.

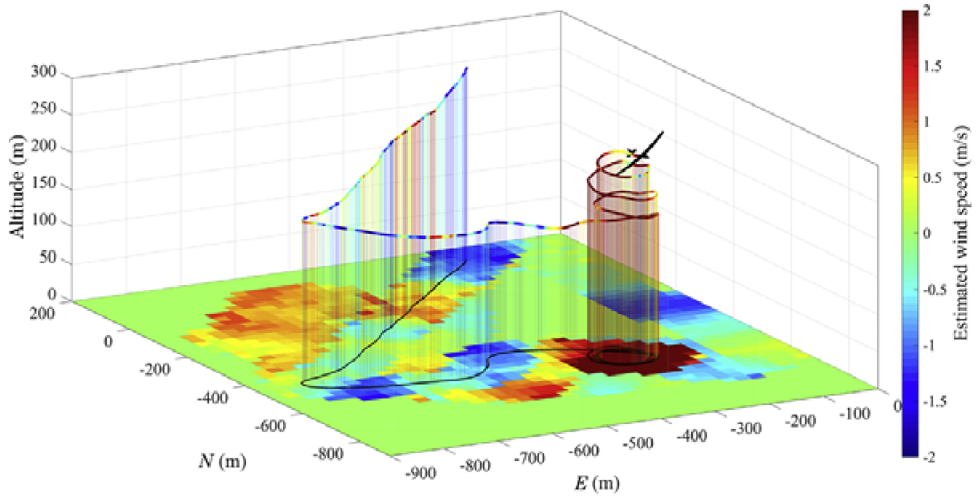
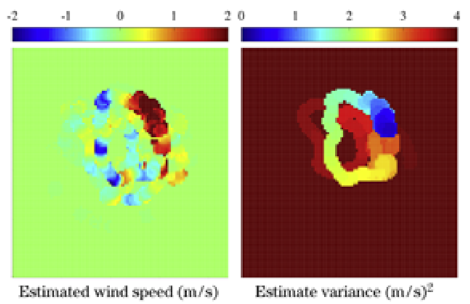
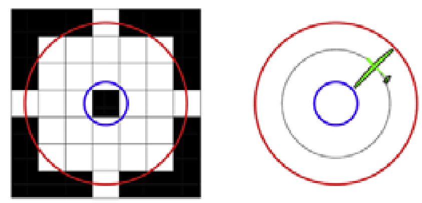


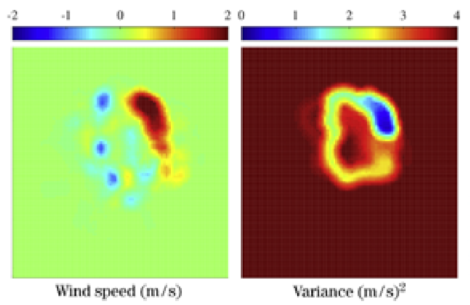
FIGURE 5 An example showing how wind measurements taken during flight are incorporated into the energy map. The regions distant from the flight path shown were visited earlier during the flight, but within a thermal decay time. The aircraft is shown enlarged for visibility. The plot shows actual flight data



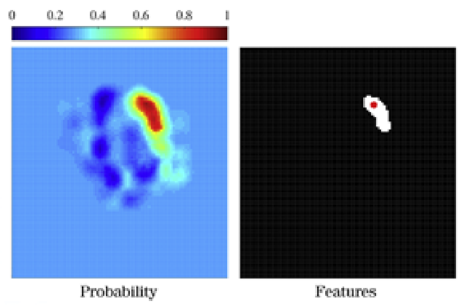
(a) Unprocessed estimated wind energy map and the associated estimate variance. Blue regions in the variance map represent areas where the aircraft recently visited and trusts the estimated wind speeds.



(b) Kernel used in the convolution approximates the area crossed during a typical circling climb. The red circle is the expected maximum orbit radius, the blue circle the minimum.



(c) Convolved wind and variance maps showing the expected conditions in a climb centered at each cell. These maps are the result of the convolution of the maps in Figure 6a with the kernel in Figure 6b.



(d) The probability that a climb centered at each cell exceeds a minimum threshold wind speed is plotted on the left. A minimum probability threshold is applied to this map and features identified on the right. The strongest expected wind during a climb in each feature (a peak in figure 6c) is indicated by a red dot.

FIGURE 6 The process by which discrete lift features are segmented from the wind map. The map is treated as an image, so that standard image processing techniques (e.g., convolution) can be used

3.4 | Identifying features in the wind map

Because the wind map holds estimates of the wind speed at the center of each discrete cell and ignores correlations between cells, further processing of this "raw" map is required to extract useful features for decision making and navigation purposes. An example map of estimated wind speeds and the associated variance map are given in Figure 6a.

3.4.1 | Identifying lift

To extract energy from a feature, the lift must be large enough to circle within. Sufficiently large regions can be identified through a convolution of the wind map with a mask that represents a typical turn radius, shown in Figure 6b. The result of the convolution is a map of expected wind speeds for climbs centered at each cell (an example shown in Figure 6c). The variance map is treated similarly.

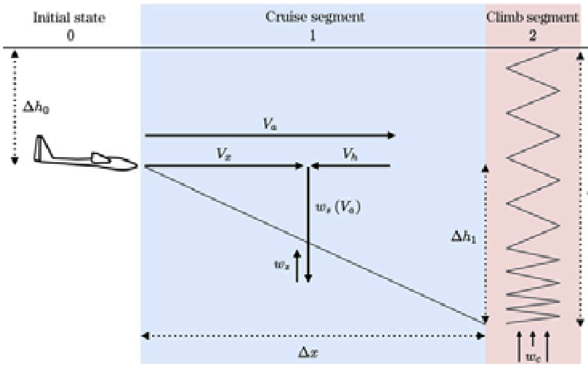


FIGURE 7 Diagram showing relevant parameters for an approximation of equivalent climb rate for a distant lift source. The same figure can be used to derive classical speed-to-fly rules

Features are extracted from the mean wind speed and variance maps by determining the probability that the lift experienced through an orbit would provide a reasonable climb rate. This is evaluated by calculating the cumulative distribution function of the Gaussian distribution described by the convolved magnitude and variance maps. In the map in Figure 6d, a high probability indicates that there is a significant chance that a climb centered at that cell will experience wind speeds equal to or greater than the threshold value.

Finally, the probability map is thresholded to yield a binary map where only distinct features remain. Cells that comprise each feature are grouped, and the size, expected strength, variance, and location of the feature are computed. Each feature can then be evaluated (as in Section 3.4.2) to determine its utility to the aircraft.

The lift feature map is evaluated every 2 s given the most recent wind energy map (containing both estimated mean and variance) available. The list of features and key properties are published to the rest of the system at the same rate.

3.4.2 | Evaluating lift

The utility of a lift source depends on its strength, distance, and the aircraft's current energy needs. Quantitatively, the value of a lift source can be defined as the expected value of $\frac{\Delta e}{\Delta t}$ for a glide to the lift followed by a climb, illustrated in Figure 7.

For a lift source i, the *equivalent climb rate*, which relates the energy lost in transit to the total climb expected in the lift source, is

$$\left(\frac{\Delta e}{\Delta t}\right)_i = \frac{\Delta e_{\text{cruise to } i} + \Delta e_{\text{climb in } i}}{\Delta t_{\text{cruise to } i} + \Delta t_{\text{climb in } i}} \tag{16}$$

On arrival at the lift source, the aircraft will circle, achieving a climb rate w_c. For simplicity a 30° nominal bank angle is assumed, with airspeed selected to minimize sink rate per Equation 2 with $\phi = 30°$. The expected climb rate can then be expressed as

$$w_{c,i} = w_{z,i} - w_{s,\min}(30°) \tag{17}$$

where $w_{z,i}$ is the expected strength of that thermal obtained as in Section 3.4.1.

With the parameters defined Figure 7, the equivalent climb rate in Equation 16 can now be expressed as:

$$\left(\frac{\Delta e}{\Delta t}\right)_i = \frac{\Delta h_0 w_{c,i}\left(v_a - w_{h,i}\right)}{\Delta x\left(w_{c,i} - w_s(v_a) - \frac{w_{h,i} w_s(v_a)}{v_a} - w_z\right) + \Delta h_0\left(v_a - w_{h,i}\right)} \tag{18}$$

Note that in Equation 18 as the aircraft approaches the feature ($\Delta x \to 0$), the value of that lift source ($\frac{\Delta e}{\Delta t}$) approaches the expected climb rate in that feature (w_c). This allows randomly encountered lift to be compared with lift on the map when making decisions to maximize $\frac{\Delta e}{\Delta t}$. Lift sources that cannot be reached are assigned a climb rate of 0.

During flight, the aircraft constantly updates its evaluation of each of the mapped lift sources identified in Section 3.4.1. The aircraft also assesses the expected arrival altitude, the probability that the thermal can be safely reached, and the probabilities of violating terrain clearance limits en-route to or on arrival at the lift source. This more detailed evaluation of each lift source is published every 5 s.

3.5 | Lift occupancy map

An occupancy grid mapping approach is employed separate from the wind map to keep track of locations where lift is frequently found, modeling the relative likelihood that lift will be found in a given region. This is distinct from the wind energy map, which estimates the current state of the environment. The lift occupancy map provides long-term memory, much like human pilots who identify "house thermals" that repeatedly appear in the same location (Reichmann, 1978); here it is an approximation of the likelihood that lift can be found in a cell.

The lift occupancy map is a set of $M \times N$ two-dimensional cells similar to the wind energy map. If m_{ij} is the cell ij in the lift occupancy map, its state can be either occupied by lift x_{ij} or not (denoted $x_{ij}^{\neg}$). Given a measurement, $z_{ij,k}$, taken near a cell, the inverse sensor model represents the probability that $z_{ij,k}$ indicates that lift is present in cell ij:

$$p(x_{ij}|z_{ij,k}) = 1.0 - p(\mathcal{N}(w_{z,k}, R_{ij,k}) \leq w_{z,\min}) \tag{19}$$

Because a single visit to a cell provides relatively little information about the prevalence of lift, a low weight is applied to updates to the occupancy grid map. This forces the map to evolve slowly and prevents the map from quickly saturating. Similarly, negative measurements do not necessarily imply that lift can never be found at a location, thus a lower weight is assigned to negative than positive indications.

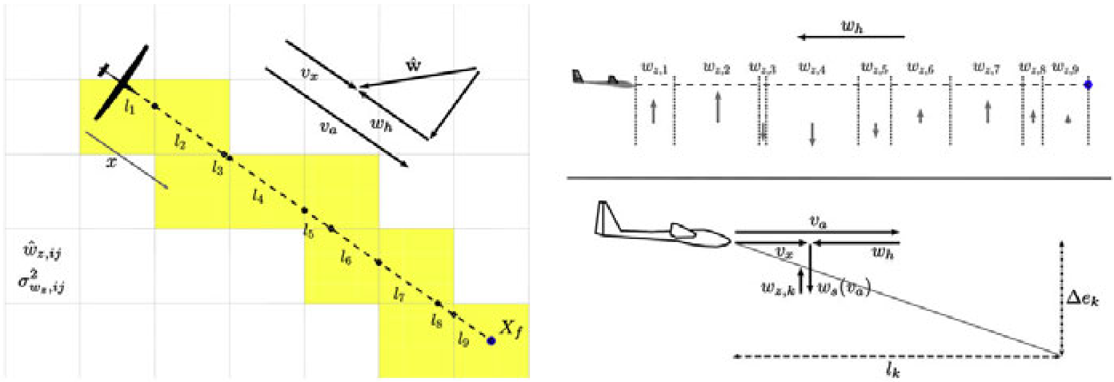
The log odds cell update is then

$$\ell_{ij,k} = \alpha \log \frac{p(x_{ij}|z_{ij,k})}{1 - p(x_{ij}|z_{ij,k})} \tag{20}$$

$$\alpha = \begin{cases} 0.0045 & \text{if } p(x_{ij}|z_{ij,k}) \geq 1/2 \\ 0.0009 & \text{if } p(x_{ij}|z_{ij,k}) < 1/2 \end{cases} \tag{21}$$

where a negative measurement carries 1/5 of the weight that a positive one does.

The lift occupancy map is updated every time a new wind measurement is available. At the end of each flight the lift occupancy map is



(a) A top view showing the set of cell crossings that comprise a path.

(b) The top figure shows the wind experienced by the aircraft, the bottom figure is a depiction of the longitudinal kinematics of an individual cell crossing.

FIGURE 8 An illustration of a flightpath crossing multiple cells in the wind map. The portion of the flight leg that lies within each cell is indicated

saved to disk. This allows the aircraft to learn the locations of thermal producing regions over a series of flights. This map of occupancy values will be used in later development to guide exploration of the environment toward regions where the aircraft has historically encountered lift.

4 | GUIDANCE AND CONTROL

4.1 | Glide range and energy prediction

The atmospheric energy map presented in Section 3.3 contains the aircraft's best estimate of the wind conditions. The vertical wind in each cell is approximated by a random variable with mean $\hat{w}_{z,ij}$ and variance $\sigma^2_{w_{z,ij}}$. This allows us to predict the environment's impact on the energy state of an aircraft traversing the map.

$$\dot{e}_{\text{total}} = \dot{e}_{\text{drag}} - w_z + \dot{e}_{\text{motor}} \tag{22}$$

Assuming gliding flight, $\dot{e}_{\text{motor}}$ can be eliminated. The anticipated energy state of the aircraft on arrival at a destination location X_f can be found by the following expression:

$$e_{X_f} = e_0 + \int_0^{X_f} \dot{e}_{\text{total}}/v_x \, dx \tag{23}$$

Because the atmospheric energy map is divided into discrete cells, a flight from one location to another can be represented as a connected set of cell crossings (Figure 8). Siddon's method is used to determine which cells are entered and l_k, the length of travel through each cell (Siddon, 1985). Here all paths are straight lines, but the methodology is not restricted to only straight segments.

For simplicity, it is assumed that the wind in each cell is constant throughout and that transients when passing between cells are negligible (as depicted in Figure 8b), thus cell crossings can be addressed independently. For a path that passes through N cells, the energy state of the aircraft at the end of the path is equal to the initial energy state e_0 summed with the changes due to each cell crossing.

$$e_{X_f} = e_0 + \sum_{k=0}^{N} \Delta e_k \tag{24}$$

Likewise, if each cell crossing is assumed independent, the probability that the aircraft completes N cell crossings is

$$P_{X_f} = \prod_{k=0}^{N} P_k \tag{25}$$

where P_k is the probability that an aircraft which is known to be aloft before crossing cell k is still aloft after crossing that cell. Once the aircraft energy state is propagated through a cell, the probability that the route up to that point can be traversed is the cumulative probability of successfully crossing each of the prior cells independently. For the route in Figure 8, the probability that the entire traversal is successful is the cumulative probability of all N crossings.

The simplified kinematics of an aircraft crossing an arbitrary cell k in the wind map is shown in Figure 8 b. The aircraft can be expected to choose an airspeed $v_a(w_{z,k}, w_h)$ that minimizes the energy loss through the cell crossing, described in Section 4.2). The time required to cross a cell is

$$v_{a,k} = f(w_{z,k}, w_h) \tag{26}$$

$$\Delta t_k = \frac{\Delta x}{v_x} = \frac{l_k}{v_{a,k} - w_h} \tag{27}$$

and the change in energy due to crossing a single cell can be expressed as

$$\Delta e_k = \Delta t_k \left(w_s(v_{a,k}) - w_{z,k} \right) \tag{28}$$

With perfect knowledge of the wind across the domain, propagating an aircraft's state along the path in Figure 8 would be a deterministic operation given by Equation 24. However, since the updraft environment is uncertain another stochastic approach is taken, where an aircraft's energy state is affected by a random process in each cell crossed.

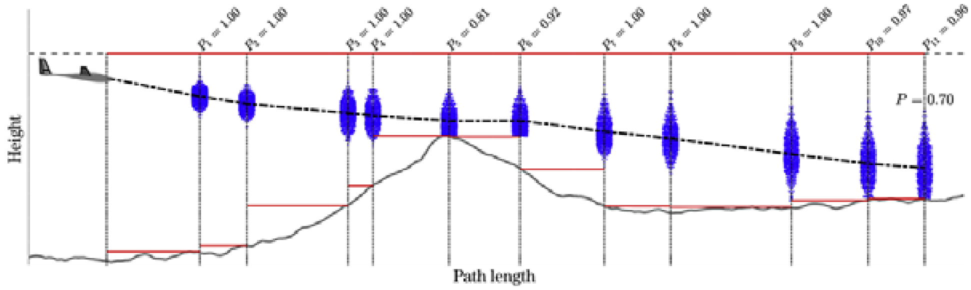


FIGURE 9 A numerical approach is taken to approximate the energy cost of flight to an arbitrary destination. Each cell boundary crossed is shown by a vertical dotted line. The particles representing possible aircraft states following each resampling step are shown by the blue distributions. The minimum $h_{k,\min}$ and maximum $h_{k,\max}$ allowed altitude in a cell are shown by red lines. The expected altitude throughout the transit is depicted by the dashed line

The wind map is used to obtain the estimated values of $w_{z,k}$ and its variance, $\sigma^2_{w_{z,k}}$, while the drag model in Equation 2 is used to approximate energy lost. What results is a probability density function (pdf) that describes the distribution of possible states that could result as the aircraft traverses the uncertain representation of the environment in the map.

Mission and environmental limits impose nonlinear constraints on the state distribution in this problem, making analytic determination of the aircraft state pdf difficult. A particle method is used to numerically evaluate Equation 24, an example of this approach applied to an arbitrary path crossing a terrain feature is illustrated in Figure 9. For particle i of a set of M, an individual cell crossing is evaluated by

$$w_{z,k,i} \sim \mathcal{N}\left(\hat{w}_{z,k}, \sigma^2_{w_{z,k}}\right) \tag{29}$$

$$v_{a,k,i} = f(w_{z,k,i}, w_h) \tag{30}$$

$$e_{k,i} = e_{k-1,i} + \Delta e_{k,i} \tag{31}$$

where $\Delta e_{k,i}$ is evaluated as in Equation 28 given the vertical wind speed $w_{z,k,i}$ sampled from the mapped distribution. The airspeed $v_{a,k,i}$ is selected given the sampled vertical wind speed and expected headwind w_h. The energy states of all M particles are projected through a single cell crossing in this manner before the distribution is resampled.

Before the next cell crossing is evaluated, a new set of candidate energy states are sampled from the particles that survived the previous crossing. This satisfies the Markov assumption and makes cell crossings independent events, validating the probability computation in Equation 25. The independent cell crossing probability is evaluated after crossing k but before re-sampling:

$$P_k \approx \frac{\text{number of particles aloft}}{\text{total number of particles}} = \frac{\sum_{1 \le i \le M} \left[\left(e_{k,i} - v^2_{a,i}/2g\right) > h_{k,\min}\right]}{M} \tag{32}$$

where $h_{k,\min}$ is the minimum allowed altitude (either by terrain or mission limits) in cell k. Additionally, those candidate states that are projected to climb above the maximum altitude $h_{k,\max}$ are prevented from doing so, reflecting either environmental or mission-related constraints.

This method for evaluating flight through the environment allows for a number of properties to be approximated. The probability that the aircraft is able to reach a destination is easily computed by Equation 25. The expected energy state at any cell boundary can be determined from the particle distribution at that point, allowing the likelihood of violating an altitude clearance to be computed and providing an estimate of the expected arrival altitude. Finally, this method can be used to approximate the distance that an aircraft can fly along a given path before the probability of it violating an altitude constraint becomes unacceptably high.

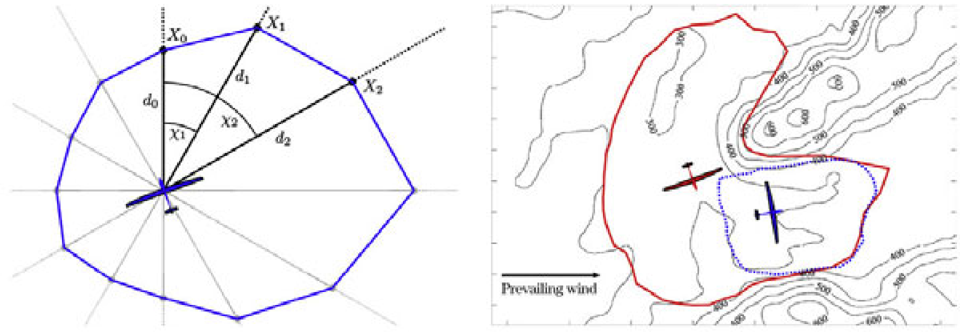
4.1.1 | Glide footprint

The approach developed in Section 4.1 can be used to compute the glide footprint of the aircraft, illustrated in Figure 10. This represents the set of locations that are reachable with high probability. This is similar to range calculations done by glide computers used by sailplane pilots; however, the glide footprint calculated here uses the wind map which contains information not available to a conventional glide computer.

The glide footprint is determined by approximating the range of the aircraft along a set of uniformly spaced bearings (χ_k in Figure 10a) from its current location. The number of bearings chosen dictates the resolution of the glide footprint evaluation. In practice, 20 bearings is sufficiently dense to approximate the glide range and yet remain computationally tractable.

To evaluate a single bearing χ_k, a path is defined extending from the aircraft's current location along that bearing. Flight along this path is evaluated as in Section 4.1 until the cumulative probability of survival given in Equation 25 falls below a set threshold at range X_k. This is repeated for each bearing, and the glide footprint is approximated by a polygon with vertices at each of the path endpoints.

The shape of this polygon is influenced by wind and terrain features. Because this method assumes straight-line flight, it does not necessarily represent the full extent of the reachable region: first, because it does not compute a path around complex topography; second, because it does not allow diversion from a given heading to exploit mapped lift. This method does however provide a computationally efficient tool for judging the feasibility of destinations while incorporating knowledge stored in the wind map.



(a) Illustration of the paths used to evaluate the glide footprint. The polygon connects the points X_k along each bearing χ_k where the probability that the aircraft can no longer continue exceeds a threshold.

(b) The glide footprint of an aircraft with an initial altitude of 600m is depicted by the solid red polygon. The footprint for the same aircraft with an altitude of 500m and at a different location is shown by the dotted blue polygon. Terrain contours are depicted in thin black lines.

FIGURE 10 The glide footprint of the aircraft is evaluated in flight given the information contained in the wind map. With a probability of 0.95 the aircraft can reach any point within the enclosed region

4.2 | Optimizing flight speed

The glide/climb cycle presented in Figure 7 can be used to develop the optimal speed to fly in between climbs if we assume that lift sources have a static location. This speed to fly is the airspeed that maximizes the instantaneous value of $\frac{\Delta e}{\Delta t}$ as expressed in Equation 18. If the sink rate w_s is represented by the quadratic speed polar presented in Equation 1, the optimal speed to fly is:

$$v_{stf} = \frac{aV_h - \sqrt{a\left(aV_h^2 + bV_h + c + w_z - w_c\right)}}{a} \tag{33}$$

The value w_c represents the next expected climb rate and the other terms are as defined in Figure 7. w_c will occasionally be referred to in this work as the “MacCready” value after the setting used on a circular slide rule developed by Paul MacCready for computing speed to fly given an estimate of w_c (the term continues to be used by sailplane pilots). If the MacCready value is zero (the aircraft does not expect to encounter lift), the expression in Equation 33 represents the speed to fly which maximizes glide range.

The resulting command guides the aircraft to fly slowly in lift to gain energy and quickly in sink to escape it. The aircraft should also fly faster into a headwind and more slowly with a tailwind. Finally, the aircraft should fly faster if it is confident that it can reach a strong lift source and quickly regain altitude, and more slowly if only weak lift is available.

Note that the optimal speed to fly given by Equation 33 is not explicitly dependent on the current state of the aircraft. An implicit altitude dependence will be introduced through the expected climb rate, w_c.

If the climb rate in the next lift source is unknown, an approximation must be made of w_c in Equation 33. This is the situation typically faced by manned and autonomous gliders without a wind map.

The strategy for determining the expected climb rate with altitude has been the subject of previous research (Cochrane, 1999b) and has been incorporated in other autonomous soaring systems (Edwards & Silverberg, 2010). This value typically decreases with altitude; at lower altitudes, the aircraft has less time to encounter strong lift before reaching its minimum altitude. When the aircraft is high, it can afford to pass up weak lift, confident that it can reach a stronger updraft.

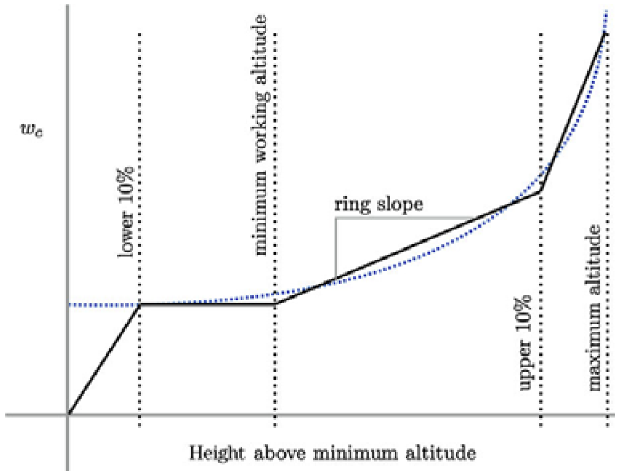


FIGURE 11 The heuristic MacCready value. This approximates the expected climb rate within strongest lift source that the aircraft should encounter given its current altitude. The blue dotted line shows the approximate shape of the MacCready function used in other work (Edwards & Silverberg, 2010)

Figure 11 shows the shape of the MacCready function, $w_{c,h}$, used in this work. The piecewise function approximates the shape of curves described elsewhere (Cochrane, 1999b; Edwards & Silverberg, 2010), the discrete segments made adjusting behavior from the ground straightforward during flight.

During flight testing, it appeared that updrafts were not as coherent close to the surface as they were at higher altitudes. There was an altitude (usually near one-third of the boundary layer depth) below which lift was significantly more difficult to find and exploit. Above this altitude, the aircraft could reject weaker thermals, but the best course of action below the critical altitude was to exploit any lift available. This effect has been observed by glider pilots and in numerically optimized MacCready schedules (Cochrane, 1999b). Including this effect causes the MacCready value to fall off rapidly at low altitudes and results in the schedule depicted in Figure 11.

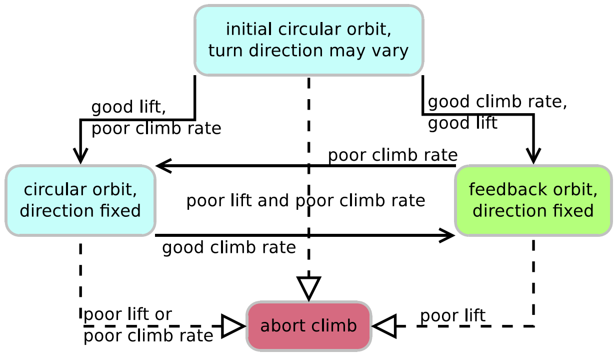


FIGURE 12 Finite states and transitions involved in the multipart climb strategy

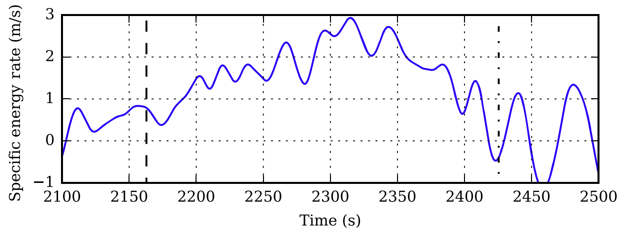


FIGURE 13 Thermalling modes. The aircraft transitions to feedback thermalling once it is confident the thermal is strong (thick dashed vertical line). The controller fine-tunes the climb, improving climb rate. As the aircraft nears the top of the thermal and climbs more slowly, it transitions back to information gathering (dash-dot line) then abandons the thermal after determining that it goes no higher. The climb rate shown is smoothed from flight data using a filter with cutoff frequency approximately equal to the circling period

In this application, the aircraft has additional information however—the lift map provides the locations of thermals of known strength. If a mapped lift source is evaluated to have a higher equivalent climb rate than the altitude based curve in Figure 11, then the aircraft should use that value for making decisions.

The MacCready value actually used by the aircraft is then dependent on both the current altitude and the state of the map.

$$\hat{w}_c = \max\left((\Delta e/\Delta t)^{*}, w_{c,h}\right) \tag{34}$$

With $\hat{w}_c$ determined, Equation 33 can be solved for the speed to fly at any time given the current wind (w_z) estimate and the headwind (V_h). A number of behavior and control decisions will also be based on the immediate value of $\hat{w}_c$.

4.3 | Thermal centering and climb control

The objective of the climbing strategy is to maximize the rate at which energy is gained from a thermal by circling within the strong core and by tracking the thermal as it drifts and evolves. The controller must trade updraft strength near the thermal center with increasing sink rate as the aircraft turns more tightly.

Previous researchers have tested methods to exploit thermals. The technique used by Allen in the first autonomous soaring flights and later used by Edwards involved simply flying circles centered on the region of strongest lift (Allen & Lin, 2007; Edwards & Silverberg, 2010). Andersson developed a feedback controller (Andersson et al., 2009), which uses energy acceleration, $\hat{\ddot{e}}$, to compute a turn-rate command. Under conditions with little measurement noise or latency, this algorithm has been proved to center a thermal (Andersson, Kaminer, Dobrokhodov, & Cichella, 2012b; Andersson, Kaminer, & Jones, 2010).

A combined approach is adopted here, with transitions implemented in a finite-state machine, depicted in Figure 12. Regardless of the climb strategy, an estimated feature center location, a turn direction, and a nominal orbit radius are needed. Immediately after "latching" into thermal mode, the center of the lift feature is estimated. The aircraft turns toward the center point at an initial turn radius selected to achieve a 30° bank angle at an airspeed that minimizes sink rate.

In flight testing, it was observed that wind drift complicated the process of finding and staying in the thermal's core. To keep the aircraft within the thermal while it gathers enough information to locate the core, only simple circles are permitted initially. This information gathering period lasts for 40 s, during which the climb will only be abandoned if mission limits are violated.

After the initial centering period expires, the aircraft has the option to abandon the thermal if climb rate is inadequate, either because the current lift source is weak or because a stronger source is known or expected to be within reach (detailed in Section 5.2.3). While thermalling, the aircraft will switch between two exploitation techniques depending on whether the mean climb rate meets $\hat{w}_c$, the expected climb rate.

If the measured lift appears suitable but the climb rate is less than anticipated, the aircraft will continue making circular orbits to gather more information about the thermal structure. If the center location is stable and the climb rate is greater than w_c, the controller switches to a mode that blends circular orbits and energy feedback similar to Andersson's (Andersson et al., 2012b) controller.

During both thermalling modes, the climb rate of each orbit is evaluated and the nominal orbit radius perturbed using the extremum seeking method developed by Daugherty and Langelaan (2014). This results in an orbit radius that slowly varies as the climb progresses and as the aircraft seeks the orbit radius that maximizes the mean achieved climb rate. The combination of thermalling mode switching and extremum seeking allows the aircraft to gather information about a thermal and fine-tune its approach to thermalling to maximize climb rate. Flight data showing this behavior are shown in Figure 13.

The aircraft begins its climb by flying a constant-radius turn ($2100\text{s} \le t \le 2160\text{s}$). This allows the thermalling controller to assess thermal strength and localize the updraft. In this case, the thermal was strong enough and the feedback controller was activated, fine-tuning the climb. As the aircraft reaches the top of the thermal, the climb rate decreases. At $t = 2425\text{s}$ the aircraft switches back to circular orbits in an effort to recenter the thermal. Here, the aircraft had climbed to the thermal top and abandoned the thermal to begin exploration.

When an estimate of the thermal center location is required, it is computed from a queue of measurements, similar to that used by both

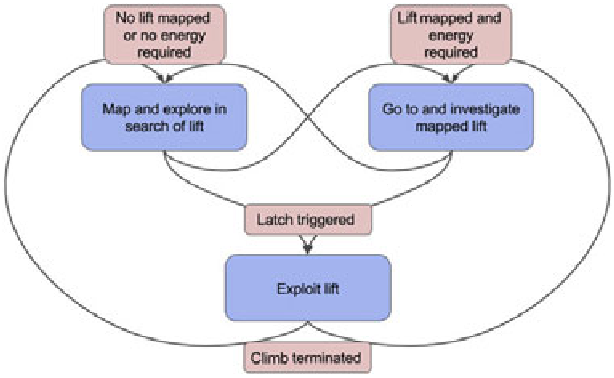


FIGURE 14 Diagram showing the primary discrete system behaviors that make up the autonomous soaring system (blue), with connected by possible transitions (arrows), and transition criteria (red). Note that the behaviors shown are high-level and in some cases contain several constituent discrete states

Allen and by Edwards (Allen & Lin, 2007; Edwards, 2008a). Upon the initiation of a "latch," the localizer computes the centroid of the queue weighted by the energy rate, similar to Allen's technique (Allen & Lin, 2007). This location is also used to initialize a nonlinear least-squares optimizer, which estimates parameters of a thermal model.

The thermal observation model uses Allen's thermal profile (Allen & Lin, 2007), modified to explicitly include the thermal position in north/east/down coordinates (referenced to the same datum used for the maps). The updraft strength and thermal location are estimated while the environmental sink and radius are assumed fixed to ensure observability.

During the initial thermal encounter, where the aircraft has only flown across the profile, short queues are used (around 15 s). Once the aircraft has obtained more measurements from a thermal, the queue size is increased to 45 seconds, approximately two turn durations.

5 | HIGHER LEVEL BEHAVIORS

Methods for understanding the environment and controlling the aircraft have been discussed. Now we cover decision-making strategies for maximizing both our knowledge of the environment and the aircraft's energy state. Aircraft actions are divided into discrete behaviors, each with a definite purpose designed to contribute directly to the mission of prolonging endurance. Thus the "behavior" of the autonomous soaring system arises through the combination of a set of relatively simple actions.

A finite state machine (shown in Figure 14) is used to manage transitions between the various behaviors required for the complete autonomous soaring system. The state machine tracks the current system state and measurements and commands a transition between discrete states when rules are triggered, conditions are exceeded, or limits are violated. States are either exploration behaviors (where new lift sources are sought), or exploitation behaviors (where known lift is utilized to increase aircraft energy). The tools previously described are utilized to determine when transitions should occur.

5.1 | Exploration

Exploration is driven by the desire to maximize utility of the map. Earlier work has focused on minimizing map variance (Lawrance & Sukkarieh, 2011). However, from the standpoint of thermal soaring, it is more useful to explore areas where thermals are likely to be found. Here an approach based on biased priority is used. The soaring environment is divided into discrete cells which correspond to those in the lift map. The center of the qualifying (is within the glide footprint) cell *ij* that maximizes

$$Q_{ij} = \text{priority}_{ij}\text{bias}_{ij} \tag{35}$$

is chosen as the destination waypoint. This concept of biased exploration priority Q_{ij} will be explored in the next two sections.

Once the aircraft reaches the assigned waypoint, a new destination is chosen in the same manner. This process continues while the aircraft is engaged in the exploration behavior. An exploration target will be abandoned (and a more feasible one chosen) if it is determined to no longer be in range (as determined by the glide footprint) or if the exploration behavior is abandoned in favor of exploiting lift. The transitions that will end the exploration behavior are the initiation of a latch onto promising lift or choosing to pursue a remote mapped lift source (see Figure 14).

5.1.1 | Exploration priority

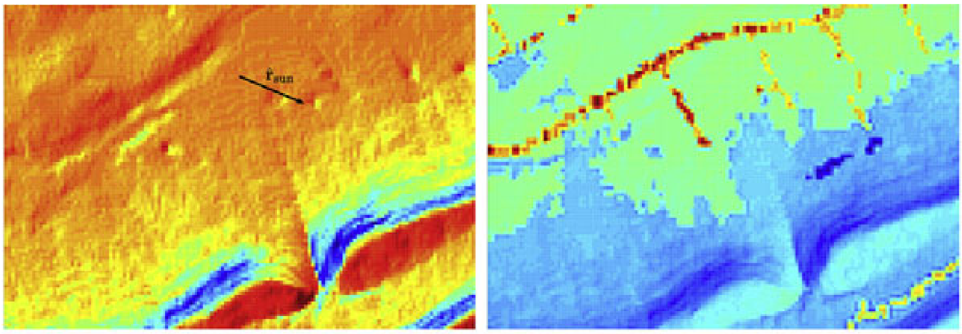
Cells are evaluated and assigned a priority value that emphasizes uncertainty in the wind map and areas likely to produce valuable lift. This focuses exploration activity on areas that are quantitatively determined to have high value and have not been recently visited. An exploration priority map is generated with cells that correspond to those in the wind energy map presented in Section 3.3.

Historical observations from the occupancy map (Section 3.5) and characteristics of the underlying terrain are used to assign priority to cells. Terrain information is specified a priori using data from the National Land Cover Database (NLCD) (Homer et al., 2015) and the National Elevation Dataset (NED) courtesy of the U.S. Geological Survey.

Because thermals are the result of heat transfer between the surface and overlying air, regions with high surface heat flux (h) receive priority. Surface heat flux depends on the local incidence angle and magnitude of incoming solar radiation, surface albedo α, and the Bowen ratio β (the ratio of sensible to latent heat flux). Expressed for the properties of a given cell *ij*, the approximate sensible heat flux (Stull, 1988) can be represented as

$$h_{ij} = \frac{-s\left(\hat{\mathbf{r}}_{\text{sun}} \cdot \hat{\mathbf{n}}_{ij}\right)\left(1-\alpha_{ij}\right)}{1+\frac{1}{\beta_{ij}}} \tag{36}$$

For an example region, the sensible heat flux is approximated and plotted in Figure 15b. Even though the insolation in Figure 15a is constant over much of the region, the varied land cover leads to differences in the expected heat flux. In particular, paved surfaces show a high heat flux, while fields appear more favorable than forest-covered regions.



(a) Normalized insolation on mapped terrain at 1100 local time in mid summer. Red indicates direct sunlight on a surface, the vector $\hat{\mathbf{r}}_{sun}$ is in the direction of the sun.

(b) Normalized approximate sensible heat flux at the surface given the insolation in figure 15a. Red indicates a high rate of heat transference to the surrounding air, here associated with paved roads.

FIGURE 15 Land cover characteristics and insolation evaluated over a sample region to show their effect on exploration priority

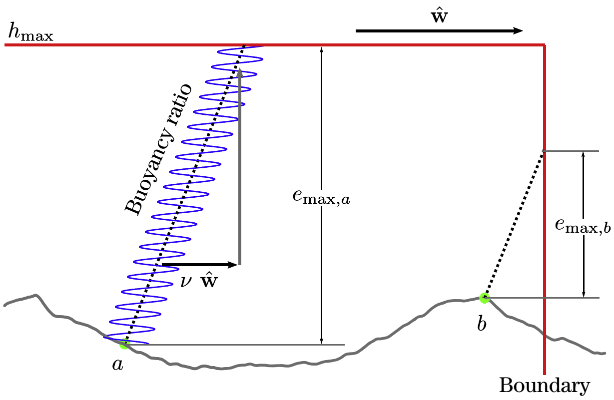


FIGURE 16 Diagram showing how the maximum energy in a region of the map is interpreted. This is used when computing the utility of exploring a region

The heat flux map is normalized so that values lie on the range [0,1] and then scaled to develop a heuristic *a priori* probability of encountering lift over the mission region, so that $P_{\text{heuristic},ij} = \alpha_h ||h||_{ij}$.

The history of wind measurements captured in the lift occupancy map discussed in Section 3.5 is also used to guide exploration. This allows erroneous assumptions about lift-producing regions to be corrected by measurement. To encourage frequent exploration of these regions as well as the implicitly attractive regions, the lift-occupancy map takes precedence when it indicates an overwhelmingly attractive area.

$$P_{ij} = \max\left(p(x_{ij}),\ P_{\text{heuristic},ij}\right) \tag{37}$$

where $p(x_{ij})$ is the lift occupancy probability for cell ij computed in Section 3.5.

The value of lift is determined by the maximum energy that could be gained in a region, which may be limited by mission requirements, terrain features, wind direction, and convection height. This is illustrated by $e_{\max,k}$ in Figure 16. The minimum allowed altitude in a cell, the prevailing wind direction, and the expected buoyancy of lift determine this value. For example, the downwind edge of the map is less attractive on a day with strong winds because the aircraft will only be able to climb for a short time before being forced out of the mission region.

Given the approximate probability of encountering lift and the value of that lift, the expectation of energy available in a cell is

$$E\left[e_{ij}\right] = e_{\max,ij}\, P_{ij} \tag{38}$$

The uncertainty of the wind estimates in a region also influences exploration. A region known with certainty not to be producing thermals should be avoided, even if it has historically produced thermals or has attractive terrain characteristics. This leads to the following definition of priority:

$$\text{priority}_{ij} = \sigma^2_{\hat{w}_{z,ij}}\, E\left[e_{ij}\right] \tag{39}$$

where $\sigma^2_{\hat{w}_{z,ij}}$ is the variance in the wind estimate described in Section 3.3.

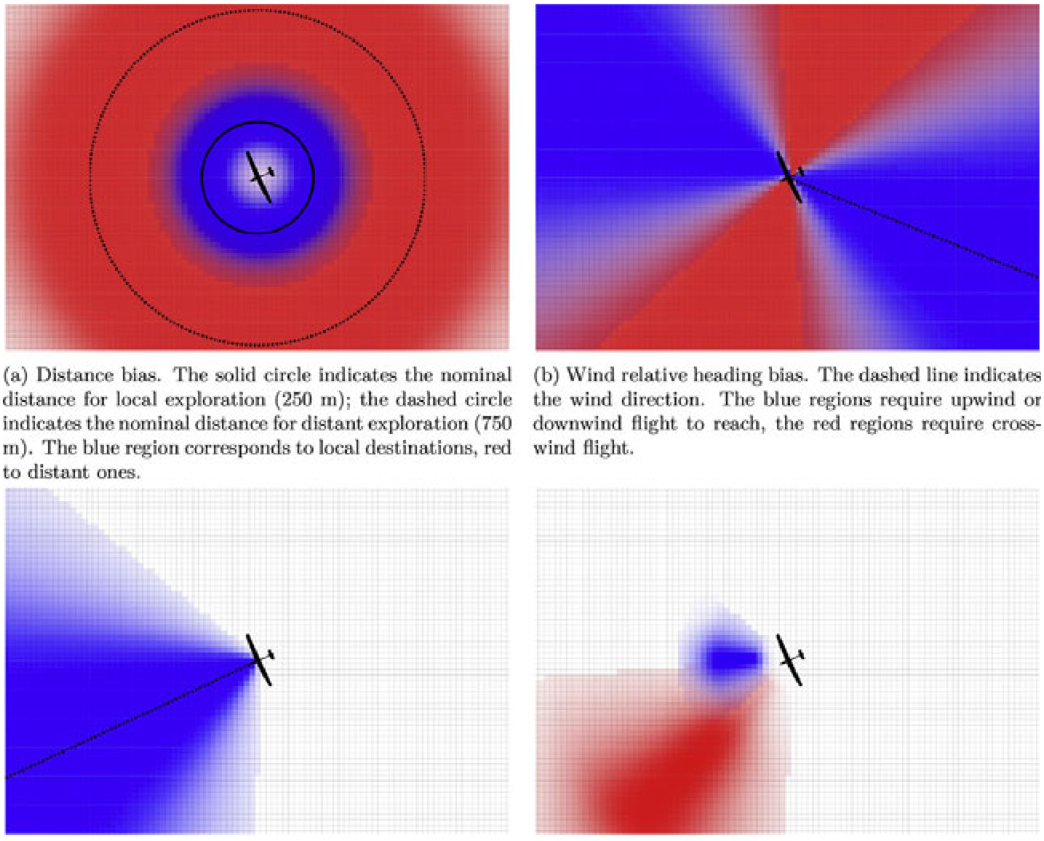
The highest priority regions for exploration will thus be areas that are expected to generate lift, have a large working altitude band, and in which the current wind estimates are highly uncertain.

5.1.2 | Exploration bias

In addition to exploration priority, biases based on aircraft heading, the current map, and current wind direction affect search strategy.

The default behavior is local exploration (where the nominal distance to a destination waypoint is 250 m, about 15 s flight time at normal flight speeds). This minimizes uncertainty in mapped wind estimates in the vicinity of the aircraft (so that mapped thermals are near enough that they can be easily reached). Local exploration is favored unless the wind map shows strong sink near the aircraft. In that case, the nominal exploration distance is set to 750 m, forcing a longer transect to a less-explored region (Figure 17a). In either scenario, the nominal distance is upper bounded at 75% of the maximum range (as determined by the glide footprint in Section 4.1.1) to ensure the destination is reachable.

Following the experience of sailplane pilots (Reichmann, 1978), exploration is biased toward upwind or downwind destinations if the aircraft is flying through lift when a new waypoint is chosen and will be biased toward crosswind destinations if the aircraft is flying through sink (Figure 17b). This behavior is justified by the observation that areas of lift and sink generally align along the wind vector. If an aircraft



(a) Distance bias. The solid circle indicates the nominal distance for local exploration (250 m); the dashed circle indicates the nominal distance for distant exploration (750 m). The blue region corresponds to local destinations, red to distant ones.

(b) Wind relative heading bias. The dashed line indicates the wind direction. The blue regions require upwind or downwind flight to reach, the red regions require cross-wind flight.

(c) Relative heading bias. The dashed line indicates the aircraft's current heading, preference is for straight line flight.

(d) Combined biases with a minimum distance constraint applied. The blue region indicates a preference for local flight upwind or downwind. The red region indicates a preference for a distant destination located crosswind.

FIGURE 17 Biases shaping the exploration priority function used to choose destinations while exploring. The intensity of the color corresponds to the weight applied to an individual cell. The aircraft is shown enlarged by a factor of 60 for visibility

is in sink, it can escape by flying orthogonal to the prevailing wind direction. If the aircraft is in lift, it should attempt to fly upwind or downwind to remain within the line of lift.

The final component of the bias favors destinations that require small course deviations to reach. The preference for straight line flight is depicted in Figure 17c. This follows the observation that turning flight requires more power than level flight (see Equation 2). Straight-line flight with minimal course deviations reduces the rate of energy loss while the aircraft is exploring.

A weighting function computed in each cell is used to enforce these desired behaviors. Each behavior is expressed by dictating a nominal state x_{nominal} and the weight applied to a cell is a function of its "distance" from that nominal state ϵ_{ij}. Given in Equation 41, this form for the bias function was chosen due to the ease of defining the width of the favorable section (through the variable m), control over the roll-off (b), and the weight that the bias carries (ν).

$$\epsilon_{ij} = |x_{ij} - x_{\text{nominal}}| \tag{40}$$

$$B_{ij,x}\left(\epsilon_{ij}\right) = \frac{1}{\left(1 + e^{-b(\epsilon_{ij}-m)}\right)^{1/\nu}} \tag{41}$$

A minimum distance constraint is additionally enforced by a barrier function of the form:

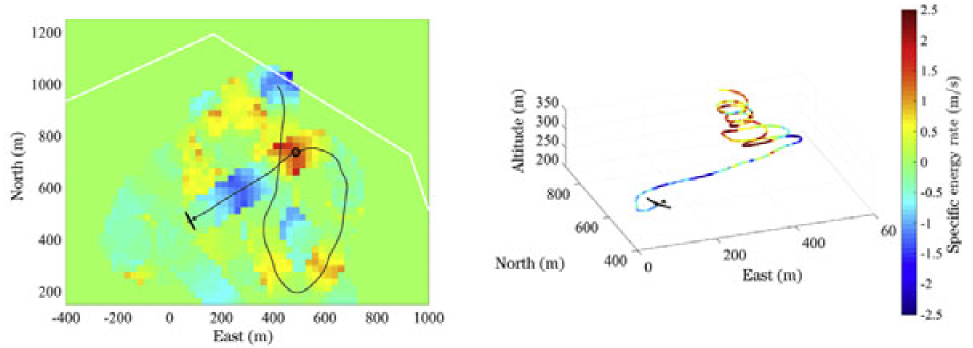
$$B_{ij,\text{minimum distance}} = \min\left(\left|\frac{x_{ij} - x_{\text{nominal}}}{x_{\text{nominal}}}\right|^{\mu}, 1\right) \tag{42}$$

The scale factors in Equation 41 and 42 were adjusted through simulation and refined during flight tests to develop the desired behavior.

The total bias applied to a cell is the product of the individually computed biases

$$\text{bias}_{ij} = B_{ij,\text{wind}}\, B_{ij,\text{distance}}\, B_{ij,\text{bearing}}\, B_{ij,\text{minimum distance}} \tag{43}$$

When a new exploration destination is chosen, the total bias is computed and multiplicatively applied to the regional priority map to obtain the biased priority in Equation 35. Figure 17d shows two examples where the bias is computed. The cell weighting depicted in blue might correspond to an aircraft that is experiencing lift and is in a region where estimates in the wind energy map are uncertain. In this case, the aircraft favors an upwind destination that is relatively nearby. The cell weighting depicted in red might show an aircraft that is very certain of local wind estimates and experiencing light sink. In this case, the aircraft picks a distant location to continue its search.



(a) Energy map at the time "go to" was triggered and a flight trace showing the preceding 180 seconds.

(b) Flight trace showing 180 seconds following "go to" trigger.

FIGURE 18 Sample of flight showing result of "go to" trigger. The specific energy rate colorbar for subfigure 'b' also applies to subfigure 'a'

5.2 | Exploitation

Because using a thermal to climb is a discrete action, the act of beginning a climb is referred to as a "latch" and departing a climb as an "unlatch." Deciding when to latch and unlatch has been a problem faced by every researcher to demonstrate an autonomous soaring system and some very capable rules have been developed to guide the decision (Allen & Lin, 2007; Andersson et al., 2010; Edwards & Silverberg, 2010). Here the energy map is used to aid latch/unlatch decision making.

5.2.1 | Deciding when to pursue mapped lift

Recall that the aircraft's expectation of its future value of specific power $\frac{\Delta e}{\Delta t}$ was defined in Equation 34 and is denoted $\hat{w}_c$. This single value represents all of the knowledge we have about the environment, both what is mapped and what we anticipate based on prevailing conditions and the current altitude. It is also the primary variable on which lift exploitation decisions are based.

The goal of the aircraft is to maximize $\frac{\Delta e}{\Delta t}$, thus it should never accept lift weaker than $\hat{w}_c$ and should accept equivalent or stronger lift when available. In the simplest case, if an aircraft happens across lift and is confident that it can achieve a climb rate greater than $\hat{w}_c$ in that lift source, it should stop to climb. It should depart any lift source that yields a climb rate lower than $\hat{w}_c$. This is how the ring setting is traditionally used by glider pilots and is similar to the latching criteria used by other autonomous soaring researchers.

Use of the ring setting in this way means that aircraft flight behavior changes due to mapped lift within gliding range: strong mapped lift in gliding range leads to more aggressive flight behavior (i.e., the aircraft does not latch onto weak thermals). This behavior will continue until the mapped lift is nearly out of reach, at which point the aircraft will turn toward known lift.

This "go to" action is triggered based on the following conditions. The first condition requires that the equivalent climb rate ($\frac{\Delta e}{\Delta t}$) expected in the best mapped and reachable lift source must be greater than the value of $w_{c,h}$ as plotted in the graph in Figure 11. This implies that the aircraft does not expect to encounter a stronger lift source than the best currently mapped and within reach if it continues to search. If this condition is met, and any of the following conditions are true, the aircraft will abandon the current action and head directly for the best mapped lift source:

1. The smallest expected altitude clearance during the transit drops below a threshold.
2. The expected altitude clearance on arrival at the location of the lift source is below a threshold.
3. The probability that the aircraft can reach the lift source drops below a threshold.

The first condition prevents the aircraft from seeking out weak or distant lift when it is likely to encounter better lift through search. The three listed conditions then ensure that the aircraft chooses to seek lift while it has enough energy to reach, locate, and begin exploiting a thermal.

If no mapped lift sources match the first condition (none are better than what the aircraft can expect to randomly encounter), or the best lift source does not fit any of the listed conditions, the "go to" action will not be triggered and the aircraft will continue its current behavior.

While en route to mapped lift, the aircraft will stop to climb if the latch conditions discussed in Section 5.2.2 are met. If the aircraft reaches a mapped lift source's location but does not encounter the expected lift, a large orbit (120 m radius) is commanded with the center point at the mapped location of the lift. This maneuver is used to find a thermal that has moved or was inaccurately mapped. If this clearing orbit is completed and no lift is encountered, the aircraft will have gathered enough measurements to erase the erroneously identified lift from the map.

Flight data showing the result of a "go to" trigger are shown in Figure 18. Figure 18a shows the energy map at the time "go to" was activated as well as the flight trace of 180 s preceding the "go to." The aircraft found a lift source located at (450, 750) but initially did not latch because it was not strong enough given the aircraft's altitude at that time. It subsequently flew through strong sink and realized there was no other lift in range (and it was unlikely to encounter a stronger source by chance), and it decided to return to the lift source. Figure 18b shows a flight trace of 180 s following the "go to" decision (colored by rate of change of specific energy). The aircraft arrived at the mapped thermal location above the arrival clearance altitude of 200 m and initiated a search; it found the thermal slightly north of the mapped location (winds were blowing out of the southwest) and initiated climb where it ultimately gained approximately 300 m altitude.

5.2.2 | Thermal detection

Regardless of whether a lift source was sought intentionally or encountered inadvertently, the aircraft should only stop to circle if it is confident that it is crossing a lift source large enough to support a climb. To eliminate smaller features such as gusts from triggering a latch, a filtering process is applied to a queue of prior wind speed measurements, and thresholds are chosen to reduce the impact of these features.

Other authors have used time-based filtering of energy rate (or vertical wind speed) measurements; here we use a metric based on distance flown in lift. This distance-based approach effectively checks the diameter of a lift source while being robust to head- and tail-winds.

The threshold climb rate $\hat{w}_c$ is corrected for the expected aircraft sink rate in thermalling configuration (30° bank and tracking the airspeed that minimizes sink rate at that bank angle (see Equation 2) to determine the minimum wind speed $w_{z,\min}$ that will support climb at $\hat{w}_c$:

$$w_{z,\min} = \hat{w}_c + w_{s,\min}(30^\circ) \tag{44}$$

This minimum lift strength is the threshold with which measured winds are compared when deciding when to latch and unlatch.

In contrast to the mapping and energy estimation strategies presented earlier in this work where no time history is explicitly maintained, exploitation decisions are made based on a finite queue of vertical wind and climb rate measurements. For this purpose, a 45-s long queue of these measurements is kept, along with the location and time at which each measurement was taken. The information in this queue can then be processed in an efficient manner that allows for informed latch and unlatch decisions to be made.

The criteria for determining whether a latch should be initiated is computed as follows: The vertical wind speed measurements in the queue are first thresholded against the minimum acceptable lift $w_{z,\min}$ such that the only information preserved is whether each measurement exceeds the threshold or not. This has the effect of reducing the impact of large measurement outliers, a quality that was found to be necessary during flight testing. The alternative strategy utilized by autonomous soaring researchers is to take a weighted mean of the climb rate measurements in the time history; however, this can be directly affected by large outliers (Allen & Lin, 2007; Edwards & Silverberg, 2010).

After the threshold operation, a simple dilation is applied to the resulting binary wind speed queue. This operation serves to eliminate short periods of sink from affecting the prevailing trend. Finally, the distance along the aircraft's flightpath over which the resulting binary queue continuously indicates good lift must exceed a threshold to trigger a latch. This threshold distance is chosen to approximate the size of a suitably large thermal, here a distance of 80 m is used.

$$\text{latch IF}\{w_{z,k} > w_{z,\min}\},\ \ \forall k \in \text{previous 80 m} \tag{45}$$

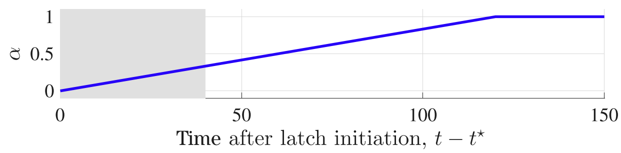


FIGURE 19 The threshold reduction during the initial phase of thermal centering immediately following a latch. The lockout period (where the climb will not be abandoned regardless of measured lift) is shown by the gray region. Afterwards, the full threshold is gradually applied

5.2.3 | Thermal unlatch

Ideally a climb continues until reaching the mission ceiling, but in some circumstances the vehicle should discontinue a climb for other reasons. The aircraft should depart a lift source if the climb rate is inadequate, if a sufficiently stronger lift source becomes available, or if the aircraft has climbed to an altitude where it expects to find stronger lift elsewhere. In all of these cases, the decision variable $\hat{w}_c$ as discussed previously is the threshold used to trigger the transition. Put simply, a climb is abandoned if the time history of the achieved climb rate $\hat{\dot{e}}$ shows a lower rate of actual energy gain than $\hat{w}_c$.

As an aircraft climbs, the achieved climb rate must meet a higher threshold to continue thermalling ($\hat{w}_c$ increases with altitude). Additionally, if the map is well-populated with thermals the aircraft will climb in a weak lift source only until it has enough altitude to reach a stronger one. If the aircraft is climbing within the strongest mapped lift source, only the altitude component, $w_{c,h}$, will influence the decision to unlatch.

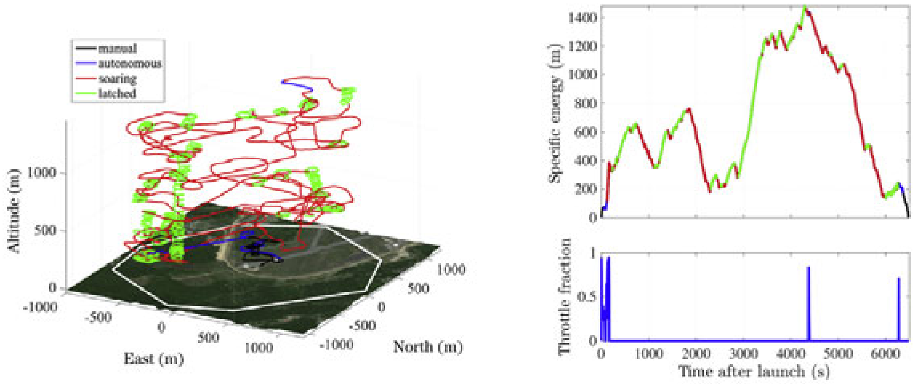
It often takes several orbits for the center estimation algorithm to settle on a feature center. This is common to human pilots, who often investigate a thermal for several turns before identifying the core (Cochrane, 1999a). Additionally, it often takes several orbits for the extrema seeking algorithm to determine the best radius at which to orbit the estimated thermal center. It is thus expected that the aircraft will experience a reduced climb rate at the beginning of a climb.

Hence state transitions are disallowed (except for safety reasons) for the first 40 s after a latch to provide the aircraft enough time to find the thermal center. Following this lockout, a scaling function slowly brings the climb rate requirement up to $\hat{w}_c$ (Figure 19). The relaxed climb rate requirement is given:

$$\{\hat{\dot{e}}_k > \alpha(t_k - t^\star)\hat{w}_c\},\ \ \forall k \in \text{40 previous seconds} \tag{46}$$

To evaluate the transition criteria for ending a climb, the climb rate measurements during the prior 40 s of flight are compared to the scaled threshold as in Equation 46. If more than half of the measurements during this time exceeded the weighted climb rate threshold, then the criterion is satisfied and the aircraft remains in the thermal.

This process will cause an aircraft to abandon a climb if the achieved climb rate does not match $\hat{w}_c$. However, the reaction time to periods of significant sink is slow. In some cases, the aircraft should move on if the thermal falls apart quickly or if the core is not tracked. Hence an



(a) Example flightpath drawn showing system modes. Several long climbs are depicted as well as short explorations of possible lift. Note differences in land cover shown in the satellite photo (taken from Google Earth).

(b) Corresponding barogram and throttle command plots.

FIGURE 20 Flight profile of a typical soaring flight. The flight trace is colored by flight mode; the barogram shows specific total energy as a function of time, and the throttle plot shows throttle setting. The motor was used only to aid launch and recovery and was briefly engaged after loss of communications to (autonomously) fly the aircraft to the lost link waypoint (where communications were regained and the soaring system was reengaged)

additional thermal rejection criterion based on the minimum required vertical wind speed ($w_{z,\min}$ discussed in Section 5.2.2) is used:

$$\{w_{z,k} > \alpha(t_k - t^{\star})w_{z,\min}\}, \ \forall k \in 20 \text{ previous seconds} \tag{47}$$

If greater than half of the wind speed measurements during the preceding 20 s of flight fail this threshold, then the climb is abandoned.

Both criteria are monitored continuously throughout a climb, if either is failed after the initial lockout period expires, the aircraft will immediately move to abandon the current lift source (as shown by the climb state diagram in Figure 12). After a lift source is rejected, the aircraft is again faced with the decision presented in Section 5.2.1 and shown in the state transition diagram in Figure 14. If another mapped lift source does not meet the requirements for triggering a "go to" action, the aircraft will pursue an exploration behavior, otherwise it will go to the best mapped lift source.

5.2.4 | Thermal climb evaluation

During a climb, the performance of the aircraft relative to the threshold climb rate $\hat{w}_c$ is used to select the exploitation strategy. The simple circling strategy is used until enough information about the thermal has been gathered to enable a strong climb with the feedback technique. The selection metric is given:

$$\{\hat{e}_k > \hat{w}_c\}, \ \forall k \in 20 \text{ previous seconds} \tag{48}$$

If greater than half of the climb rate measurements during the preceding 20 s of flight pass this threshold, then the feedback technique is utilized.

Note that no relaxation of the threshold is used in this instance because this criterion is not used to terminate a climb. Rather, it is a way of checking whether the aircraft has found a sufficiently strong region of the thermal to climb in and is ready to enter a more aggressive energy exploitation mode. If the achieved climb rate fails this criterion, the aircraft will resort to the more conventional circular orbit exploitation mode. These transitions are shown in the finite-state machine diagrammed in Figure 12.

6 | DISCUSSION

Flight results are discussed in detail in Depenbusch et al. (2017). To aid the brief discussion of results presented here, a flight trace, barogram, and commanded throttle from a typical flight are shown in Figure 20. This flight took place at Phillips Army Airfield, Aberdeen Proving Ground, Maryland, in mid to late afternoon in September under conditions dominated by synoptic high pressure. Winds were approximately 2 m/s from the southwest, the boundary layer was capped by scattered cumulus clouds at approximately 1500 m.

The flight was limited to an area of 3.17 km^2 with an absolute ceiling of 3,050 m above mean sea level (the white polygon in Figure 20a defines the flight area). In practice, altitude was limited by clouds and air traffic control clearance; further, the aircraft was required to remain in visual line of sight at all times. Terrain at the test site was generally flat, with tree lines surrounding the airfield providing the largest changes in elevation. Land cover included paved runways and taxiways, a gravel road, mown grass, swampy scrubland, and deciduous forest. The differences in land cover result in differences in surface heat flux; these differences in heat flux affected exploration behavior and the locations where thermals could be found. Observations of bird soaring behavior (mainly vultures, but also eagles) as well as where the aircraft found lift showed that thermal locations varied by time of day: early in the day thermals triggered from pavement and along tree lines; later in the day, thermals could be found over swampy areas that warmed more slowly.

Operational modes are indicated by the coloring of the flight trace and barogram. The aircraft is in manual mode (black trace) only for launch and landing. It is in autonomous mode (blue trace—waypoint tracking though the Piccolo autopilot, with waypoints defined in a

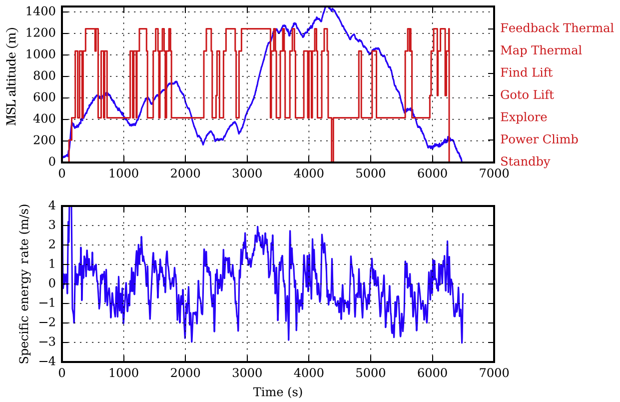


FIGURE 21 Detailed autonomy modes. When at high altitude, the aircraft would opportunistically encounter strong lift before needing a climb but as it descended later in the flight it sought out mapped lift

parameter file) prior to engaging the soaring system (red and green traces). When the autonomous soaring system is engaged, all commands originate from the on-board payload computer. The red trace indicates active soaring flight in exploration mode; the green trace indicates an active latch in a thermal. Note the brief period of nonsoaring flight (blue) at 4400 s: the aircraft flew into the ground station antenna null and lost the ground station communication link. This caused the soaring system to disengage (although full autonomy was maintained—there was no human intervention) and the autopilot commanded constant altitude powered flight to the lost link waypoint. At the lost link waypoint, communication with the ground station was automatically regained and the soaring system reengaged.

Note the long thermalling climb from 300 m above ground level (AGL) to 1300 m AGL in addition to the several shorter thermal climbs. The aircraft explored a significant fraction of the available area, driven by the exploration priority. There were also several times when the aircraft flew toward (and found) previously mapped thermals and latched into thermalling mode.

Between $t \approx 5{,}000$ and $t \approx 5{,}100$, the aircraft is briefly climbing in a thermal. Given the aircraft's altitude and the thermal strength, it departs to begin exploration. It begins a long descent, encountering a (weak) thermal at $t \approx 5{,}500$. It briefly climbs in this thermal but then departs because it is too weak given the aircraft's specific total energy (500 m). It detects and maps some weak thermals, and at $t \approx 6{,}000$ returns to a previously mapped thermal (which is now strong enough to be desirable given the aircraft's specific total energy of under 200 m). The aircraft climbed nearly 100 m in this thermal (and was continuing to climb), but it was late in the day and we decided to end the flight. During the long descent between $t \approx 5{,}100$ and $t \approx 6{,}000$, the flight region was largely covered in sink, and the aircraft was unable to find an area of strong-enough lift. This can occur when the scale of convective cells is on the same order as the diameter of the flight region (in this case, ≈2 km) (Sullivan & Patton, 2011).

Figure 21 details state transitions (upper plot) and rate of change of total energy (lower plot) during the flight depicted in Figure 20. The soaring system is activated at $t \approx$ 150s and the aircraft begins exploring. For much of the flight, the aircraft encounters lift opportunistically while it builds a map and determines the strength of the thermals available (note the mode changes from "explore" directly to "map thermal"). At $t \approx 4{,}400$ s, the soaring system is briefly disabled ("standby") when the communications link with the ground station is lost; it is reestablished and the "explore" mode is resumed. During most of the flight, the aircraft was able to find strong lift (i.e., lift that exceeds the strongest mapped lift) opportunistically. The "go to" behavior occurs four times: at $t \approx 2{,}500$ s, at $t \approx 5{,}950$ s, $t \approx 6{,}100$ s, and at $t \approx 6{,}200$ s. In each case, the aircraft is rather low, so the likelihood of finding a stronger-than-mapped thermal within the glide footprint is small. In each of the four cases, "go to" is succeeded by "map thermal." In the first case (at $t \approx 2{,}500$ s) the aircraft fails to find a thermal on its initial mapping turn and the "explore" behavior is triggered. The aircraft quickly found a thermal (possibly the originally mapped thermal had drifted away) and began climbing, gaining 200 m altitude. Climb rate decreased, triggering "map thermal" and then the "explore" behavior; the aircraft then found a strong thermal that resulted in a climb of nearly 1000 m. The next three cases all resulted in the mapped thermal being found and short climbs. Referring to the lower plot of Figure 21 and again to Figure 20 b, the weak thermal encountered at $t \approx 5{,}500$ results in brief thermalling with a climb rate of roughly 1 m/s, which is too low given the aircraft's altitude of 500 m. The later climb at $t \approx 6{,}000$ began with a low climb rate but later peaked at 2 m/s. At $t \approx 6{,}250$ s the soaring system was manually placed in standby and the aircraft was manually recovered.

7 | CONCLUSIONS

We have presented a system by which a small aircraft may extend its endurance through the use of soaring flight to extract energy from the environment. This autonomous soaring system includes thermal detection, thermal latching and delatching, thermal centering control, mapping, exploration, and flight management, covering all phases of flight except for launch and recovery.

In contrast to previous autonomous soaring developments, the aircraft maintains a global map of updraft and downdraft locations in the surrounding airmass. This map provides context to the environment in which the aircraft operates and mimics atmospheric dynamics. Additionally, the available map informs transitions between exploration and energy harvesting behaviors, allowing a transition to happen when the aircraft needs energy and can still reach a known source.

Further, by building a long-term memory of thermal producing regions, the aircraft can improve the chances of finding thermals during exploration. This was one part of a strategy that guided exploration toward regions that were considered to be of high priority. Other factors included the confidence with which the aircraft regarded wind estimates in a region, the expected heat flux at the surface arising from solar heating, and the terrain gradient with respect to the prevailing wind direction.

The exploitation strategy developed here built on the work of other researchers and was guided by the results of prior flight tests. A multipart strategy was used where the aircraft was allowed to orbit a probable thermal location to gather information before undertaking a more aggressive feedback approach to energy gain.

The system has been developed using insight gained through simulation and during flight testing. This has encouraged a robust design for all system components with an emphasis on low computational requirements. Flight data presented in this paper demonstrated 1.77 h flight time with the motor running for less than 1 min. In total, the autonomous soaring system has flown for 7.8 h, with the motor running for only 8.4 min.

ACKNOWLEDGMENTS

Portions of this research were funded by the Office of Naval Research under Grant N000141110656.

The authors thank Dr. Dan Edwards and Chris Bovais from the Naval Research Laboratory for their support of this project and for organizing the flight-test campaign at Aberdeen Proving Ground.

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How to cite this article: Depenbusch NT, Bird JJ, Langelaan JW. The AutoSOAR autonomous soaring aircraft, part 1: Autonomy algorithms. *J Field Robotics*. 20188;35:868–889. https://doi.org/10.1002/rob.21782

APPENDIX A: TABLES OF PARAMETERS

TABLE A1 Summary of parameters the software related to environmental mapping

Parameter	Nominal value	Rationale
Thermal radius for thermal location estimation	100 m	Several times turn radius, consistent with observations from flight tests.
Length of measurement queue for thermal localization	45 s	Approximately two turn durations.
Time between localizations	5 s	Approximately one-quarter turn, a turn from aligned with to cross-wind (or vice-versa).
Interthermal environmental sink	0.25 m/s	
Thermal drift rate as a fraction of wind speed	0.5	Heuristic frequently quoted by manned glider pilots.
Buoyancy ratio	[0,5]	Heuristic based on expected shear conditions
Maximum measurement distance, r_{max}	50 m	Approximate lower bound on the radius of a thermal
Maximum wind variance, $\sigma^2_{w_z,max}$	1 (m/s)^2	Expected variance in environment
Map cell discretization, dx	25 m	Approximate aircraft turn radius
Wind strength threshold for lift map	1.0 m/s	One standard deviation above environmental mean
Probability threshold for lift map	0.64	
Nominal bank angle, ϕ_{nom}	30°	Typical shallow bank used in soaring flight (Reichmann, 1978)

TABLE A2 Summary of parameters the software related to aircraft behavior selection

Parameter	Nominal value	Rationale
Ring slope	0.001 (m/s)/m	Increase in threshold climb rate with altitude
Minimum working altitude	400 m AGL	Heuristically identified altitude below which lift is difficult to utilize
Ring minimum	[0.3, 0.5] (m/s)	Minimum acceptable climb rate
Psuedoprobability weight factor on sensible heat flux when determining exploration target, α_1	0.5	
Psuedoprobability weight factor on terrain gradient when determining exploration target, α_2	0.2	
Local exploration distance	250 m	
Local exploration wind threshold	-0.5 m/s	Strong sink that the aircraft should avoid
Local exploration uncertainty threshold	0.75 (m/s)^2	
Distant exploration distance	750 m	
Minimum distance in lift to trigger latch	80 m	Minimum size of a lift source to allow circular orbits at a reasonable bank angle
Period following latch during which a climb will not be abandoned	40 s	Enough time for thermal center estimates to settle
Period over which the threshold climb rate to continue a climb is reduced	120 s	Enough time to center a thermal well
Window length over which climb rate must satisfy threshold to continue climb	40 s	Approximately two orbits
Window length over which measured wind must satisfy threshold to continue climb	20 s	Approximately one orbit
Window length over which climb rate must satisfy threshold to use feedback control while climbing	20 s	Approximately one orbit
Probability threshold for computing glide range	0.95	
Minimum terrain clearance requirement en route which triggers a turn to the best lift source	200 m	
Minimum terrain clearance requirement on arrival which triggers a turn to the best lift source	200 m	
Reachable probability threshold which triggers a turn to the best lift source	0.99	